

Mikha Gracia Sugiono

Backend Engineer | Go Microservices, Multi-Tenant Platforms, CI/CD

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Summary

Backend Engineer specializing in Go microservices and multi-tenant platform architecture. Built and shipped a 16-service ERP platform to production, sustaining 1,438 req/s at p95 357 ms with a 0.02% error rate. Experienced across the full delivery path — GitLab CI/CD, Docker, and Prometheus/Grafana/Loki observability — with a security foundation in OAuth 2.0, Role-Based Access Control, and ISC2 CC certification.

Education

Institut Teknologi Sepuluh Nopember (ITS)

2022 - Present

Bachelor of Computer Science (Informatics) | Expected graduation: October 2026

GPA: 3.55 / 4.00

Relevant Coursework: Software Engineering, Database Systems, Network Security, Operating Systems

Experience

Backend Engineer — Manajemen Cerdas Informasi (MCI) Laboratory, ITS

Jan 2025 - Present

Cerdas ERP — multi-tenant cooperative (koperasi) management platform / cerdaserp.com, koperasi.iimlab.id

- Architected and shipped a 16-service ERP platform (Go and TypeScript backends, 2 Next.js frontends) behind a Kong API gateway, applying Clean Architecture and a database-per-service pattern with logical references for independent per-service deployment.
- Load-tested to 1,000 concurrent virtual users at p95 357 ms and 1,438 req/s with a 0.02% error rate, cutting p95 by 79% through Redis tenant caching and Kong upstream load balancing.
- Built the Inventory service (Go, Hexagonal Architecture, 98 endpoints) with FIFO batch stock tracking, weighted-average costing, and multi-UOM conversion.
- Decoupled Point of Sale from direct Inventory database access into a REST integration with a compensating-transaction rollback path, eliminating an overselling race condition.
- Secured multi-tenant access with Kong API Gateway, OAuth 2.0 reference tokens, and RFC 7662 token introspection, enforcing tenant scoping at the gateway rather than in application code.
- Designed the Kubernetes deployment topology (Deployments, replicated pods, Services, horizontal pod autoscaling); production shipped on client-mandated single-VPS Docker Compose, so the manifests stand as the scale-out reference architecture.
- Automated delivery across 20 repositories with a five-stage GitLab CI/CD pipeline, multi-stage Docker images, and a registry-based build-once/pull-anywhere deploy model with one-click rollback, instrumented with Prometheus, Grafana, and Loki.

Software Engineer (DevOps Focus) & Laboratory Administrator

Sep 2024 - Jul 2026

Manajemen Cerdas Informasi (MCI) Laboratory, Department of Informatics, ITS

- Built and provisioned a 5-machine on-premise compute cluster from bare metal — OS installation, service configuration, networking, and GPU driver/runtime setup — comprising 3 identical RTX 5080 nodes for distributed computing, 1 Quadro RTX 8000 workstation for heavy ML workloads, and 1 Lenovo ThinkSystem SR550 rack server.
- Operated the SR550 as the lab's deployment host for Cerdas ERP and other research services, dispatching GPU-bound ML jobs to the RTX nodes as worker machines.
- Built and maintained CI/CD pipelines with GitHub Actions and Jenkins, adding SonarQube static analysis gates to the build stage.
- Developed a manufacturing ERP in Node.js/TypeScript (Portal Access and HRM modules) with role-based access control, REST APIs, structured error handling, and logging.
- Containerized Portal Access and ERP services as Docker-based microservices with RabbitMQ messaging.

Coding Instructor — Timedoor Academy

Nov 2025 - Jun 2026

- Taught programming and digital literacy to 20+ children and teenagers across 11 small-group classes (2–5 students each) using Scratch, Construct 3, Roblox Studio (Lua), App Inventor, HTML/CSS/JavaScript, and Python.
- Mentored students end-to-end through personal projects (2D/3D games, mobile applications, interactive websites), authored structured progress reports, and led Parent–Teacher Meetings to present outcomes and recommend learning paths.

Projects

VoCalendar — AI-Powered Voice Calendar App github.com/michaelgrace72/VoCalendar
Flutter, Firebase, GPT API

- Built a calendar application driven by natural language voice commands in Bahasa Indonesia.
- Integrated GPT for schedule parsing and Firebase for real-time sync across devices.

Secure File Encryption Application github.com/michaelgrace72/Data-encryption-Flask
Python, Flask, PostgreSQL

- Implemented an AES/DES/RC4 file encryption service with role-based access control and activity audit logging.

Credit Card Approval Prediction System github.com/michaelgrace72/CreditCardApprovalPrediction
Python, Flask, CatBoost, Docker

- Built and containerized a machine learning service for real-time credit approval risk assessment.

Technical Skills

Languages: Go, Python, TypeScript/JavaScript, Java, C/C++, Dart, SQL

Architecture & Backend: Microservices, Clean Architecture, Hexagonal Architecture, REST APIs, Database-per-Service, Multi-Tenancy, Node.js, Flask

API Gateway & Security: Kong API Gateway, OAuth 2.0, Token Introspection (RFC 7662), Role-Based Access Control (RBAC), Data Encryption (AES/DES/RC4)

Containers & CI/CD: Docker, Docker Compose, GitLab CI/CD, GitHub Actions, Jenkins, SonarQube, Kubernetes (design-level)

Observability & Testing: Prometheus, Grafana, Grafana Loki, cAdvisor, Node Exporter, k6 Load Testing

Databases & Messaging: PostgreSQL, MySQL, Redis, MongoDB, RabbitMQ

Infrastructure & Systems: Linux, Bare-Metal Server Provisioning, GPU Compute Infrastructure (NVIDIA RTX/Quadro), Network Configuration, Azure, Terraform (personal projects), Git, Postman

Certifications

ISC2 Certified in Cybersecurity (CC) 2023
Cybersecurity fundamentals, risk management, network security, and incident response.

Additional Experience

Technology Mentor — Schematics Bootcamp, “Python Programming for Beginners” 2023
Science Tutor — Middle School 2021 - 2022

Languages & Interests

Languages: Indonesian (Native), English (Professional)

Interests: Cybersecurity Research, Cloud Architecture, Open Source, Audio Engineering (production and mixing)